Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (93 sites, 395 site-years) · deadtrees v2-2

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

ΔRMSE = RMSE(with D) − RMSE(without D)

GPPsat (µmol m⁻² s⁻¹)

68% improved | 68% improved | 68% improved | 65% improved

2.5 · 0.0 · -2.5 · -5.0 · -7.5

NEPmax (µmol m⁻² s⁻¹)

74% improved | 58% improved | 76% improved | 71% improved

2.5 · 0.0 · -2.5 · -5.0 · -7.5

ETmax (mm d⁻¹)

46% improved | 41% improved | 46% improved | 44% improved

0.01 · 0.00 · -0.01

WUE (g C mm⁻¹)

57% improved | 56% improved | 57% improved | 58% improved

0.5 · 0.0 · -0.5

Sites (sorted by ΔRMSE within each panel)

Relative disturbance
site maximum, 500 m
(breaks: mean ± 0.5 SD)

- Low (< 21.6 %, 35 sites)
- Mid (21.6–34.5 %, 35 sites)
- High (> 34.5 %, 23 sites)